APU
UCDF2407ICT(DI)

DIAGNOSING TOMORROW

Predicting Diabetes Risk Through Data

Group 17

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Insights and Recommendation available.

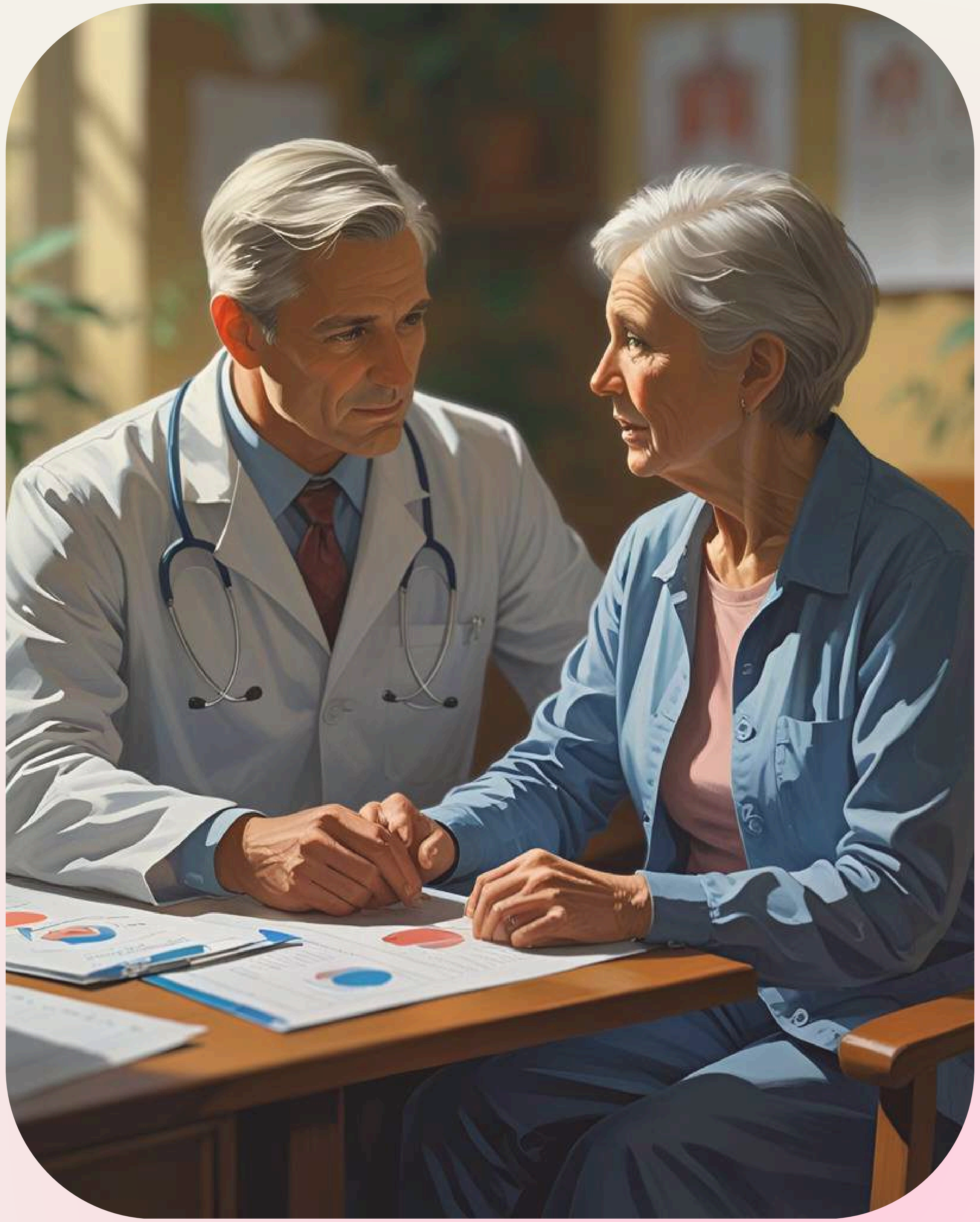
A Machine Learning Approach to Classifying Diabetes Risk Using Health Indicators

WHAT IS THIS PROJECT ABOUT?

PREDICTIVE ANALYTICS FOR DIABETES RISK USING HEALTH INDICATORS

We developed machine learning models to classify individuals as diabetic or non-diabetic using **medical and lifestyle data.**

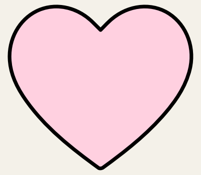
The goal: Support **early intervention** and **smarter** triage in clinical environments.



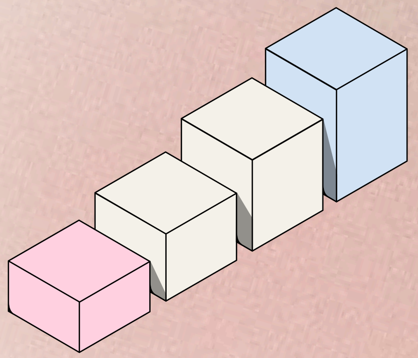
WHY THIS MATTERS ?

Our drive.

Diabetes affects over 537 million adults globally, with projections rising sharply by 2030 (International Diabetes Foundation, 2021).



Early detection reduces complications like neuropathy, cardiovascular disease, and kidney failure.



Predictive models help screen patients before symptoms escalate—saving lives and resources.

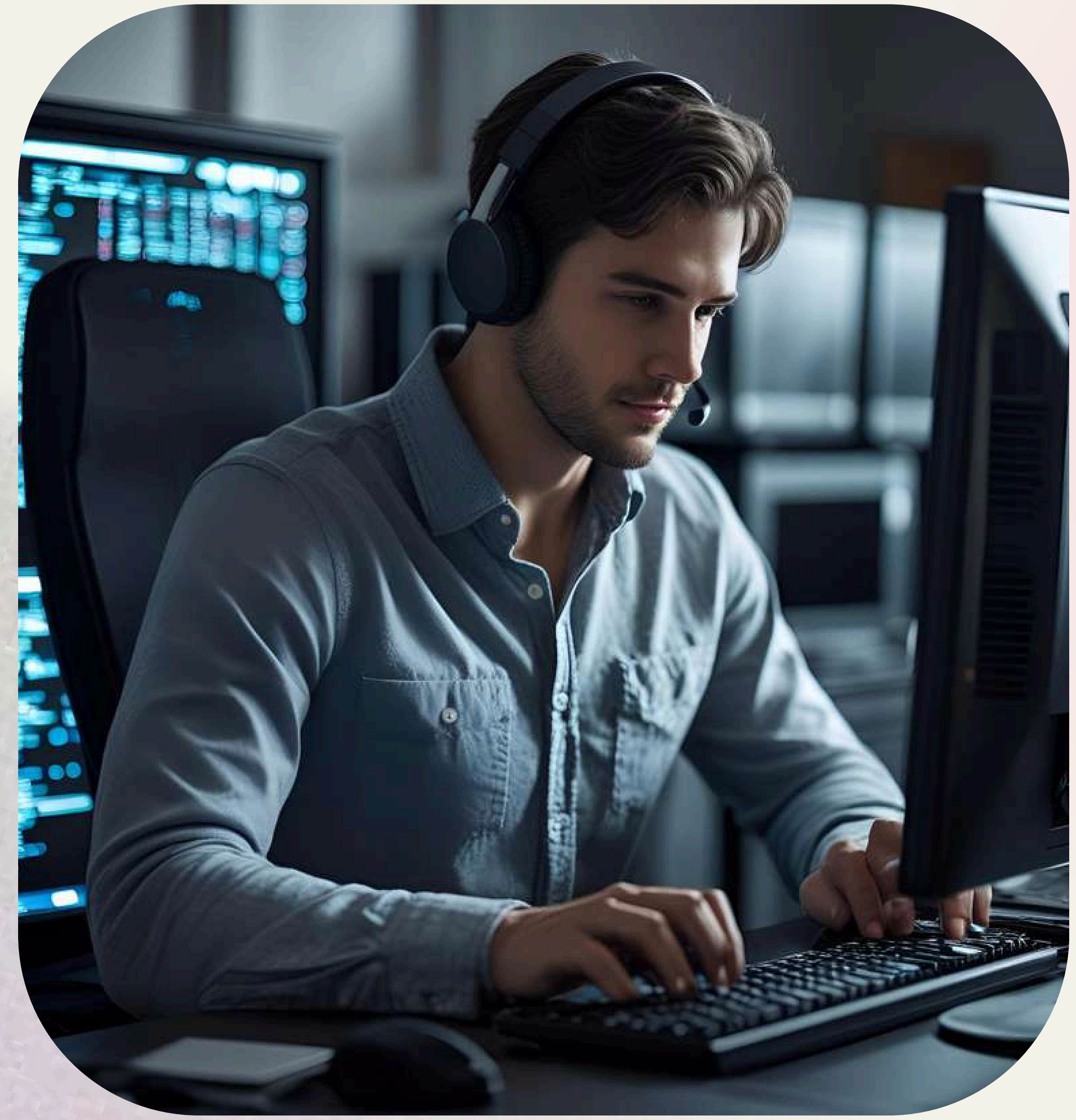
OBJECTIVES

Our goals.

- Clean and analyze the Diabetes Health Indicators Dataset
- Identify high-impact features for diabetes classification
- Train and evaluate Logistic Regression, Gradient Boosting, and Random Forest
- Recommend a model (Random Forest) for deployment in clinical workflows
- Provide data-driven insights for healthcare policy and intervention

Dataset Overview

Our source.



Name: Diabetes Health Indicators Dataset (BRFSS, 2015)

Source: Kaggle – Alex Teboul

Sample Size: 253,680 records

Target Variable: Diabetes_binary (0 = Non-diabetic, 1 = Diabetic)

Features: 21 medical and lifestyle indicators (e.g., BMI, HighBP, GenHlth, Age)

METHODOLOGY

We used CRISP-DM.

🔍 Data Understanding

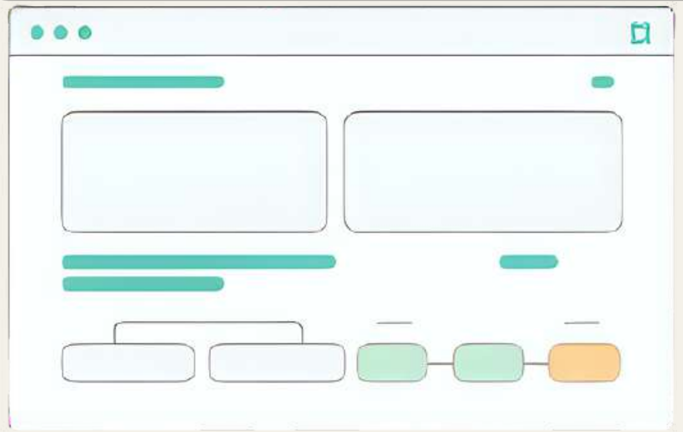
EDA revealed strong separation in BMI, Age, and GenHlth

🧹 Data Preperation

Winsorization applied to BMI, and SMOTE used to balance classes

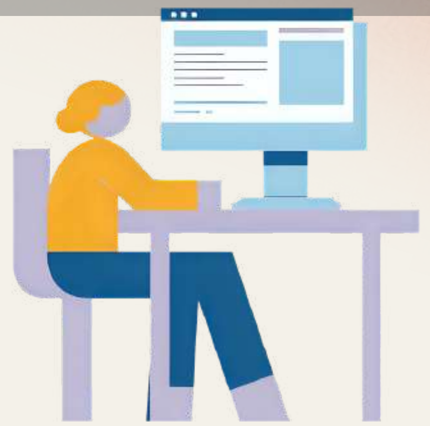
⚙️ Data Modelling and Evaluation

- Trained Logistic Regression, Gradient Boosting, and Random Forest
- Compared models using Accuracy, Recall, Precision, F1, ROC AUC



📲 Deployment Ready

Chose Random Forest for Streamlit interface

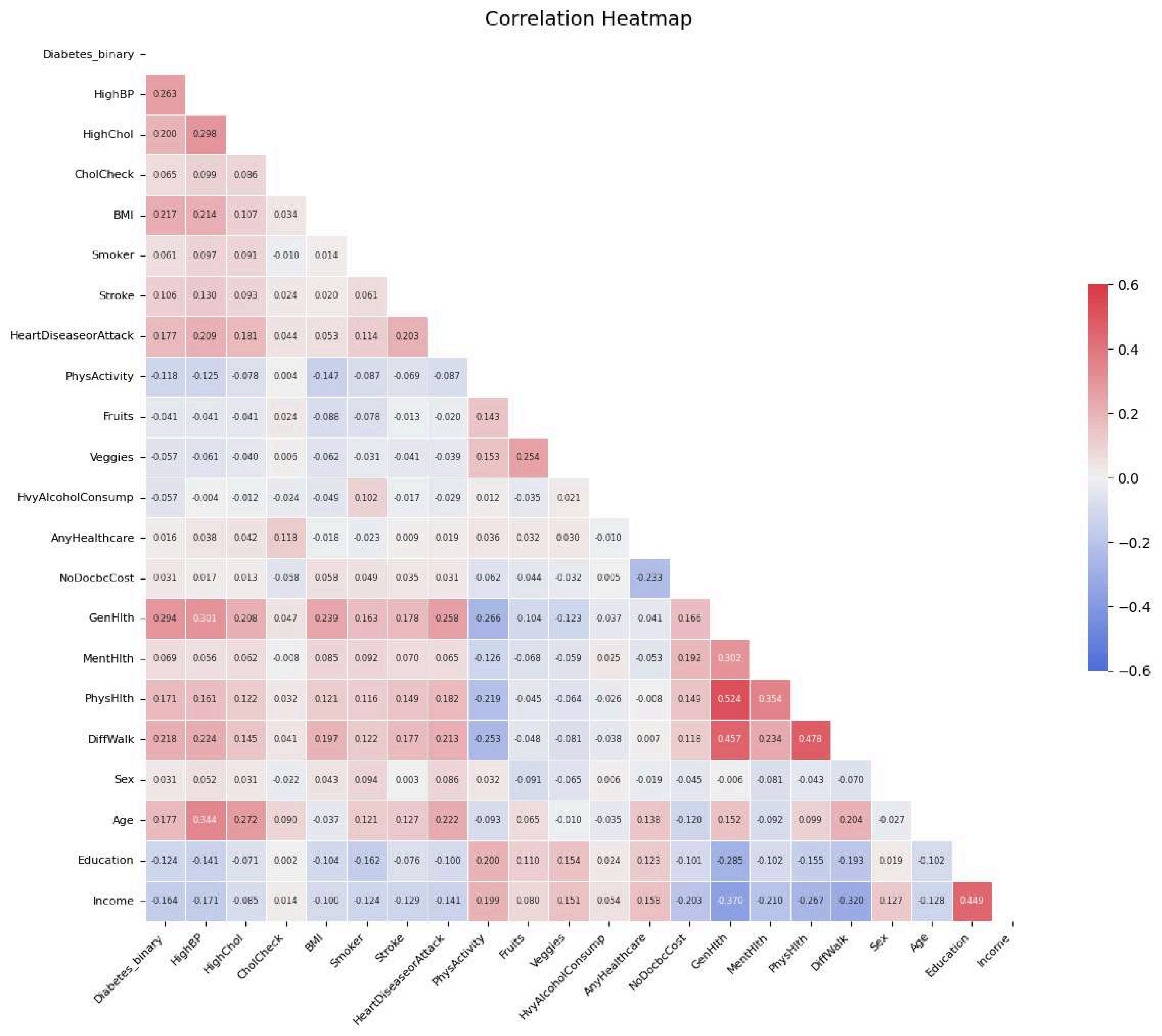


EDA (EXPLORATERY DATA ANALYSIS)

Exploration of the dataset and features.

Key Insights:

- Diabetes vs. BMI: Higher BMI correlates with increased diabetes risk
- Boxplot showed noticeable right skew among diabetic group
- Age Distribution: Diabetes prevalence rises sharply after 40
- Histogram revealed a clear upward trend in age brackets
- GenHlth Indicator: Strong link between poor general health and diabetes
- Stacked bar chart showed diabetic respondents reported lower GenHlth



Correlation Heatmap

FEATURE SELECTION

Selecting Features For Implementation.

1) BMI
2) GenHlth
3) Age
4) HighBP

Features Selected

1) Select K-Best
2) RFE (Recursive Feature Eliminiation)
3) Tree Based Elimination

Techniques Used

After comparing the feature selections, we grouped all the similar features in all algorithms used and then found proof to backup the findings

Explanation

RESEARCH

What researchers say about the features.

BMI

- Liang et al., 2025 — Global trends suggest rising obesity rates are directly driving early-onset type 2 diabetes.

- Collaborators, 2023 — Emphasizes that the global prevalence of diabetes is closely tied to obesity increases.

Age

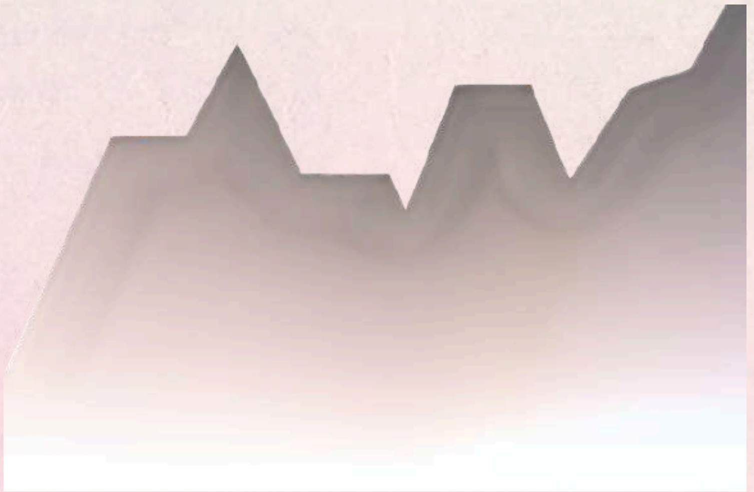
- Liang et al., 2025 — Also used to support age as a risk factor, noting higher diabetes prevalence among older age groups.

HighBP (High Blood Pressure)

- Wang et al., 2025 — Cited to justify HighBP as a critical comorbidity, with high prevalence in diabetic patients and clinical implications.

MODEL TRAINING AND EVALUATION

What algortithms we used to train and deploy.



Logistic Regression

Major Findings:

- **Has good recall but suffers in precision making it a weak competitor and will mostly overfit the positives**
- **Less suitable for clinical data and needs very high levels of tuning to make it work well**

Random Forest (Chosen)

Major Findings:

- **Random Forest offered the best balance between accuracy, precision, recall, and ROC AUC**
- **Most well rounded amongst the 3 selected algorithms**

Gradient Boosting

Major Findings:

- **Gradient Boosting showed slightly higher recall and F1-Score**
- **It required more careful tuning and was computationally more intensive**

INSIGHTS

What we found.

Top Predictors Identified: Diabetes risk is strongly associated with high BMI, poor self-rated health (GenHlth), older age, and high blood pressure (HighBP).

Class Imbalance Handled: SMOTE was applied to counter the imbalance (13.93% diabetic cases), improving fairness in model training.

Clinical Relevance:

- Predictive models can aid early screening and reduce unnecessary testing.
- Feature importance findings align with established medical literature and public health strategies.

HEALTHCARE RECOMMENDATIONS

What we recommend.

Deploy Random Forest in Clinical Settings

- Chosen for its balanced accuracy (77.01%) and strong ROC AUC (0.8371), ideal for early diabetes risk screening.

EHR Integration

- Embed the model into Electronic Health Records to automatically flag high-risk patients during routine visits.

Community Health Outreach

- Use interpretable models (like Logistic Regression) for awareness campaigns promoting lifestyle changes.

Preventive Care Prioritization

- Focus intervention programs on patients with elevated BMI, hypertension, and poor self-reported health.

Support Public Health Policy

- Align recommendations with global findings—healthy weight, improved diet, physical activity, and hypertension control can reduce type 2 diabetes risk by up to 80% (Lundgrin et al., 2025).

ETHICAL CONSIDERATIONS

Things to consider.

Data Privacy

- Recognize the importance of protecting sensitive health data in any deployment context.

Algorithmic Bias Mitigation

- SMOTE was used to correct class imbalance and ensure fairness in prediction.

Model Interpretability

- Emphasize understandable features (e.g., GenHlth and HighBP) to support patient trust and healthcare adoption.

Responsible Deployment

- Prioritize transparency, explainability, and clinical validation before integrating models into real-world systems.

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THANK YOU FOR LISTENING!

We’re open for Q&A!